



UPES ACM & ACM-W

100 Days of Code

Day: 2

Problem 1

Remove Element (LeetCode #27)

<https://leetcode.com/problems/remove-element/>

Description

A problem that strengthens understanding of in-place array manipulation by removing all occurrences of a specified value without using extra space.

Given an integer array and a target value called `val`, the objective is to remove all occurrences of `val` from the array in-place and return the count of remaining elements. The order of elements that are not removed does not need to be preserved.

This is a classic interview problem that emphasizes pointer-based techniques and in-place modification strategies.

This problem helps build concepts like:

- in-place array modification
- two-pointer technique
- index tracking
- space-optimized traversal

which are fundamental for solving array manipulation problems efficiently.

Input

An integer array and a target value `val`.

Your Task:

Remove all occurrences of a given value from the array in-place using an optimized approach and return the number of elements remaining.

Constraints: $0 \leq \text{nums.length} \leq 100$, $0 \leq \text{val} \leq 100$

Output

Print the count of remaining elements.

Sample Input 1	Sample Output 1
Array = [3, 2, 2, 3] val = 3	2 (remaining array: [2, 2])

Sample Input 2	Sample Output 2
Array =[0, 1, 2, 2, 3, 0, 4, 2] val = 2	5 (remaining array: [0, 1, 3, 0, 4])

Sample Input 3	Sample Output 3
Array = [1], val = 1	0 (remaining array: [])

Problem 2

Find the Largest Element in an Array

Description

A problem that introduces array traversal techniques and teaches how to efficiently track the largest element while processing all values in an array.

Given an array of integers, the task is to find the maximum element present in the array.

This problem appears frequently in interviews and real-world applications like ranking systems, analytics processing, monitoring platforms, and data evaluation tasks. The challenge involves maintaining the largest value efficiently during a single traversal while handling negative numbers and edge cases.

This teaches:

- array traversal
- comparison logic
- variable tracking
- optimization techniques

which are essential for efficient data processing and interview problem solving.

Input

An array of integers.

Your Task: Find the largest element in an array using an optimized approach.

Output

Print the largest element in the array.

Sample Input 1	Sample Output 1
Array = [12, 35, 1, 10, 34, 1]	35

Explanation

35 is the largest element in the array.

Sample Input 2	Sample Output 2
Array = [10, 5, 10]	10